

COS 135 Individual Assignment 6

Due: Friday 03/13/26 End of the day

~~Due: Tuesday 03/09/26 End of the day~~

This assignment has 4 programs to be developed.

Warning:

Programs that produce gcc compiler warnings or errors will receive a significant grade reduction.

What to submit:

- .c source code per each task.
- Your programs must produce similar outputs, as given, if the same inputs are provided.

[10 pts] Comments are required in the following locations (in each C source code):

- At the top of the source code comment your name and a short program description.
- Comment the purpose of each variable.
- Comment major sections of code such as input, processing, and output.

Program Design: Your program is a professional document and must be neat and easy to read. All programs should follow these specifications.

- Comments should be aligned and entered in a consistent fashion
- Blank lines should be added to aid readability
- Code within blocks should be indented
- Comments should not contain spelling mistakes
- Variable names should be meaningful

Write C programs for following tasks and submit your source codes (you must submit .c files **without compilation warnings or errors**). Sample Program inputs are highlighted in **yellow**.

Use the -Wall switch in gcc to enable all warnings:

```
$ gcc hello.c -o hello -Wall
```

(a) (10pts) Write a C program to find the sum of all even numbers between **n** and **m**. User has to input two numbers **n** and **m**, and **the program has to verify the following conditions:**

- both **n** and **m** are valid numbers and are greater than 0
- the second input number has to be greater than the first input number

hint: `scanf()` function returns the number of characters read from `stdin`.

<u>Sample output #1:</u>	<u>Sample output #2:</u>
Enter two numbers larger than 0: 23 100 Sum of even numbers between 23 and 100: 2318	Enter two numbers larger than 0: 2 K Invalid input

```
nimesha@cos135:/media/cos135/a6_codes$ sudo gcc hw6a.c -o hw6a -Wall
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: 23 100
Sum of the even numbers between 23 and 100: 2318
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: 4 12
Sum of the even numbers between 4 and 12: 24
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: 2 K
Invalid input.
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: -1 5
Invalid input.
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: R
Invalid input.
nimesha@cos135:/media/cos135/a6_codes$ ./hw6a
Enter two numbers larger than 0: 2
w
Invalid input.
```

(b) (20pts) The Fibonacci sequence is a series where the next term is the sum of previous two terms. The first two terms of the Fibonacci sequence are 0 followed by 1 (*hint: you may save these two starting values in two variables at the beginning*).

The Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144..... N

Write a program to output the Fibonacci sequence up to a given number by a user. Your program should verify the user's input as a positive integer.

<u>Sample output #1:</u> Enter a number: 90 Fibonacci Series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89	<u>Sample output #2:</u> Enter a number: -9 Invalid input - please try a positive integer.
---	--

```
nimesha@cos135:/media/cos135/a6_codes$ sudo gcc hw6b.c -o hw6b -Wall
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: 80
Fibonacci Series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: 150
Fibonacci Series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: 590
Fibonacci Series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: -100
Invalid input - please enter a positive integer.
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: 0
Invalid input - please enter a positive integer.
nimesha@cos135:/media/cos135/a6_codes$ ./hw6b
Enter a number: S
Invalid input - please enter a positive integer.
nimesha@cos135:/media/cos135/a6_codes$
```

(c) (30pts) Write a C program with following functions.

1. a C function to print all prime numbers (integers) from 1 to n
2. a C function to print all even numbers between 1 and n
3. a C function to print all odd number between 1 and n
4. a C function to print all natural numbers in reverse (from n to 1)
5. a C function to **return** sum of all natural numbers between 1 and n
(and print the returned sum inside the main() function to output)

Your program should first ask for an integer larger than 10 (i.e., $n > 10$).

Second, **validate the input**. Then, call individual functions to print output results.

Program requirements:

1. You are **NOT ALLOWED** to use global variables.
2. All functions must be **called exactly once from main()** function.
3. All functions **should have at least one parameter** to pass the user input n
4. **Functions 1-4, no return values.** Print within individual functions.
5. **Function 5 return the sum** as explained and print it in the main() function.

```

nimesha@cos135:/media/cos135/a6_codes$ sudo gcc hw6c.c -o hw6c -Wall
nimesha@cos135:/media/cos135/a6_codes$ ./hw6c
Please enter an integer greater than 10: 11
Prime numbers between 1 and 11:
1, 2, 3, 5, 7, 11
Even numbers between 1 and 11:
2, 4, 6, 8, 10
Odd numbers between 1 and 11:
1, 3, 5, 7, 9, 11
Natural numbers between 1 and 11 (in reverse):
11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
Sum of the natural numbers between 1 and 11:
66
nimesha@cos135:/media/cos135/a6_codes$ ./hw6c
Please enter an integer greater than 10: 19
Prime numbers between 1 and 19:
1, 2, 3, 5, 7, 11, 13, 17, 19
Even numbers between 1 and 19:
2, 4, 6, 8, 10, 12, 14, 16, 18
Odd numbers between 1 and 19:
1, 3, 5, 7, 9, 11, 13, 15, 17, 19
Natural numbers between 1 and 19 (in reverse):
19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
Sum of the natural numbers between 1 and 19:
190
nimesha@cos135:/media/cos135/a6_codes$ ./hw6c
Please enter an integer greater than 10: 6
Invalid input. Please enter an integer greater than 10.
nimesha@cos135:/media/cos135/a6_codes$ █

```

(d) (30pts) Write a C program to read a student's name and three grades. These three grades are passed by reference to a void function named `rank_student_grades`. This function arranges the three grades in descending order (places the largest value in the first argument, then the second and the smallest value in the third) using the swap function shown below. Finally, the program displays the student's name, grades in descending order, and the average.

→ The student's **name must be read using `fgets()`** and handled correctly if it contains spaces.

Using `scanf()` is allowed for other inputs.

→ Show maximum two decimal points.

→ You are **not allowed to change the instructions in the given skeleton program** (only to add new instructions to complete the code)

→ Global variables, other libraries, and concepts we have not covered are **not allowed**.

→ Assume always the inputs are between 0 and 100 (i.e., no need to error check the user input)

Follow the skeleton program in the next page as well as attached as a .c file with the assignment.

```

void swap(int *x, int *y)
{
    int temp;
    temp = *x;
    *x = *y;
    *y = temp;
    return;
}

```

Sample inputs:

Enter student's name = **James Bond**

Enter student's first grade = **86**

Enter student's second grade = **78**

Enter student's third grade = **92**

Sample output:

Original grades: James Bond 86 78 92

Ordered grades: James Bond 92 86 78 85.33

```

nimesha@cos135:/media/cos135/a6_codes$ sudo gcc hw6d.c -o hw6d -Wall
nimesha@cos135:/media/cos135/a6_codes$ ./hw6d
Enter student's name = James Bond
Enter student's first grade = 86
Enter student's second grade = 78
Enter student's third grade = 92
Original grades: James Bond 86 78 92
Ordered grades: James Bond 92 86 78 85.33
nimesha@cos135:/media/cos135/a6_codes$ ./hw6d
Enter student's name = Bruce Lee
Enter student's first grade = 82
Enter student's second grade = 95
Enter student's third grade = 100
Original grades: Bruce Lee 82 95 100
Ordered grades: Bruce Lee 100 95 82 92.33
nimesha@cos135:/media/cos135/a6_codes$

```

This skeleton program is also attached as a .c file with the assignment.

```

/*
 * This is a skeleton program for the Assignment 6 (d)
 * You are NOT ALLOWED to change the given instructions including the function signatures.
 * @author NR
 */

```

```

#include <stdio.h>

// function signatures
void swap(int *x, int *y);
void rank_student_grades(int *num1, int *num2, int *num3);
void print_name(char text[]);

int main()
{
    char name[20];
    int g1, g2, g3;

    // read name and three grades
    // then save them in the variables

    // example output: Adam James      68      87      79
    printf("Original grades: ");

    // call rank_student_grades() function

    // example output: Adam James      87      79      68      78.00
    // need to add the format specifier to output average
    printf("Ordered grades: ");

    return 0;
}

/*
change the values in num1, num2, and num3 in descending order
*/
void rank_student_grades(int *num1, int *num2, int *num3) {
    // complete the comparison and rank
    return;
}

/*
swap the values by reference (i.e., based on memory addresses)
*/
void swap(int *x, int *y) {
    int temp;
    temp = *x;
    *x = *y;
    *y = temp;
    return;
}

/*
print name in the array without the new line character
*/
void print_name(char text[]) {
    // complete the printing name
    return;
}

```